CHAPTER 9
Human Capital Theory: Applications to Education and Training

Answers to End-of-Chapter Questions

1. (LO1, 2) In the broadest sense, capital is defined as something that is used to make something else, and is thus not consumed (like a raw material or an intermediate input would be) in the production process. Capital does depreciate (wear out), however, as it is used, and thus it loses value over time if it is not maintained. Because capital is used as a means to an end, it is expected to generate a flow of economic returns in the future. Since there is no such thing as a free good, these returns are associated with a cost, which is usually in the form of an upfront investment. One usually expects to recoup the initial cost of the investment only gradually.

   In theory, and often in practice, human capital meets all of these conditions. It is used to produce a flow of goods and/or services from the worker, from which both parties benefit. Workers don't necessarily want the training and/or the education as an end in and of itself. They expect it to pay off in the form of higher earnings in the future. In order to create the human capital, a substantial upfront investment must be made, both in terms of explicit costs of training and education and in terms of implicit costs, such the trainees' time.

2. (LO1, 2) Given the confines of the question, we are not going to consider the intrinsic value of your education. Your profound, heartfelt love for learning, your deep intellectual curiosity, your thirst for more knowledge, your hunger for more depth, your desire to sharpen your skills — all in pursuit of that most fascinating of disciplines, namely economics — will not be considered here. Nor do we consider the satisfaction that you will receive in your future jobs from applying economics. Only the pecuniary aspects of your university education are discussed in this question. On the cost side, one considers the explicit and the implicit costs of acquiring a university education. These costs include tuition, books, and other ancillary expenses that would not be incurred if one had not chosen the university route. One would include only the portion of room and board expenditures that exceeds what one would have paid by not going to university. Recall that, had you not elected to attend university, you would still have had to eat. The implicit costs include the earnings that one would have earned otherwise. The benefits include the difference between the earnings that one receives once one leaves university and the level that one would have received over this period had one not gone to university. In order to compare the cost figures, which are normally spread out over two to four years, and the benefit figures, which are normally spread out over 40 years or so, one needs the interest rates that are paid on the borrowed money and the interest rates that would be paid on savings. The present discounted value of the costs is compared to the present discounted value of the benefits.

3. (LO5) One can assume that the training involved is specific to the firm; otherwise, the employer would not pay at all. The basic approach is to consider the costs and the benefits

from the firm's perspective with and without the training program. What wage stream is paid to the worker with and without the training? What is the marginal product stream of the worker with and without the training? This step typically involves the evaluating the expected values of the worker's marginal product period by period. Are there any explicit costs that the employer must incur, such as paying the instructor and diverting her efforts and attention from her normal duties? Worker morale might improve through training activities, but that is hard to evaluate quantitatively. It is important to know how long the training period will last, as well as how long the worker will remain at the firm after being trained or re-trained. Finally, an important factor is the probability of losing a worker (particularly through a quit) who has undergone the training. See the discussion surrounding Figures 9.6b and 9.6c of the text.

4. (LO4) For each of these programs, the most important question to ask is whether workers and firms would somehow provide and utilise these human resource programs if left to their own devices. As the textbook puts it, are there situations in which private unregulated markets do not provide a socially optimal amount of training, mobility assistance, labour market information, health and safety on the job, etc.? The textbook then goes on to give a detailed explanation of the case of publicly-funded training to workers. There are many instances in which the case for government intervention is weak. In those cases in which the trainees cannot borrow or otherwise finance any job-related training, however, the case for government subsidies is stronger (we call this situation a liquidity constraint). Many economists favour government-funded training for the disadvantaged, although in practice these programs have not always proven to be very effective. In the presence of spillover (or positive external) effects, when the training activities for one group have a favourable impact, the training services tend to be under-provided without a government subsidy, so some intervention may be warranted in such as case. The mobility, labour market information, and health and safety functions are not treated in this chapter. The basic idea is that ideally one could obtain reliable estimate of the social rate of return on investing in these various government programs. The costs of allocating resources to them should be easier to estimate. One should then allocate the resources according to the marginality principle: align the marginal cost with the estimated marginal social return.

5. (LO5) This question overlaps with the previous question. The argument in favour of government subsidies in order to correct for a socially suboptimal level of human resource development (typically this means job training) is the following. When there are positive externalities to training and education activities, the providers (either the employer or a firm contracted to provide the training) and the trainees do not capture all of the benefits that the training bestows on others. These benefits that others receive are external to their decision of how much time and money to invest in the training activities. They will thus tend to under-invest in training from a social point of view, as the marginal social benefit exceed the marginal private benefit. They only respond to the latter.

   The indirect nature of on-the-job training, where it is difficult to assess exactly how much each worker contributes to the training and what the benefits are, makes it more likely that there is an under-investment in training from a social point of view. One instance where there is widespread agreement for a major government role is in training laid-off and disadvantaged workers. A modern industrialised economy is typically subjected to changes in labour demand patterns and technology, which generate structural unemployment and underemployment. These phenomena generate high social and economic costs, and it seems pretty clear that private interests, such as firms and charities, do not have the economic resources (and sometimes they do not the have incentive) to provide job training programs and counselling to these workers.

6. (LO2) Ignoring opportunity cost: the income which could be earned from working while one's time is devoted to schooling

   Failing to discount benefits: The basic idea here is that the right to receive a payment in the future is worth less than the right to receive the same amount in the present. Higher earnings which will flow in the future from an investment in human capital must be discounted for the fact that they are not being received now.

   Double counting: one cannot count the entire cost of room-and-board expenses while attending university because one would have incurred some of these expenses anyway. One should only count the differential between the food and lodging costs between the two options.

   Considering sunk cost with no alternative value: Suppose that you are trying to decide whether or not to attend university, and that in the past you have invested in acquiring lifeguard skills. You would include the income forgone from that occupation in the calculation, but you would not include what you spent to become certified.

   Ignoring a real externality: There are many external benefits from acquiring a university degree which spill over to other individuals. Almost everyone agrees that it fosters social cohesion if and when individuals from all echelons of society can aspire to higher education and obtain it. It raises the productive capacity of the economy.

   Considering a pecuniary externality: It is not apparent that this would be an error. However the pecuniary externality might be factored into the cost–benefit analysis for the individual who is affected by the original investment in human capital, so perhaps there is a risk of counting the same cost or benefit twice.

   Ignoring consumption benefits: Many people derive utility from their degrees which goes beyond strictly pecuniary terms. They benefit from parts of their education that have little to do with their careers. People tend to enjoy most of their educational experience, and the rewards from the career to follow often include good working conditions.

7. (LO5) The "virtues" of these different channels for providing training and education are not treated explicitly in the textbook, so this discussion is a bit broader. What follows is a brief contrast between them.

On-the-job training (OJT) is likely to generate human capital which is quite specific to the firm. This is sometimes referred to as vocational training. The cost of providing it is typically shared between the employer and the worker. It is often informal in nature, meaning that there is no diploma or a certification received upon completion. The decision for investment in OJT is made by both parties, and it is one of the primary factors determining the bond between the employer and the worker. The process of providing and financing the training is governed by the mechanisms illustrated in Figure 9.6 of the text. In many cases, we expect the parties to arrive at some agreement to provide at least some degree of OJT. Although there are some difficulties in assessing and assigning the costs and benefits of OJT, they are easier to evaluate than is the case for institutional education.

Institutional training and education is likely to generate general human capital. This is more broadly applicable than specific human capital. It involves the development of skills, such as numeracy, literacy, and computer literacy, that can be applied to a wide variety of functions, as well as knowledge such as civics, history, literature, etc. that are deemed very useful in Canadian society. The decision process for how much to invest in institutional training is the human capital investment decision, which is made primarily by the worker. The employer is expected to contribute nothing. There are several reasons to believe that without government intervention, the level of investment in institutional education would likely be suboptimal. There are barriers to accessing post-secondary education, in large part due to its high costs and imperfect capital markets. The returns on education are tougher to assess than is the case for OJT. Furthermore, whereas the economic function of OJT seems to be pretty clear (to directly enhance the MRP of the worker), the exact role of institutional education is less clear. That is the basic idea behind the conflicting interpretations of investment in education as a form of human capital or as a signalling device. Each of these stories yields different policy prescriptions for the government and makes efficiency analysis (are we over- or under-investing?) more complex.

8. (LO3) In the case of zero private returns, then we would expect no private investment at all, whether one is talking about the human capital paradigm or the signalling paradigm. It is hard to imagine such a case, however. In the context of the signalling approach, if a signal leads to the worker being hired, the worker has some incentive to invest in higher education. Should the government also assume some of the cost of the investment in education? Yes, it should if the social rate of return to education exceeds the private rate of return. This could still occur within the signalling paradigm even though the education that is obtained does not direct augment the productivity of the worker. The education obtained emits a signal that helps to allocate labour to its highest and best use among alternatives. It thus indirectly raises the productivity of the worker, and perhaps the productivity of other workers as well. From that channel, the social rate of return to schooling could be high and could exceed the private rate of return. In this scenario, without a subsidy afforded to higher education, there will be under-investment in it.

Answers to End-of-Chapter Problems

1. (LO1) The benefits of the university degree are the higher earnings that you will receive for the 10 years after you leave the university. In periods 5 through 15, you will receive \$4,000 more per year. This stream of payments must be discounted by the market rate of interest. In present discounted value terms, we have:

$$\sum_{i=5}^{15} \frac{4000}{(1.05)^i}$$

The costs are all incurred between periods 1 through 4. The explicit costs are \$700 in each of these four years. To these it is necessary to add the implicit costs of the \$6,000 in foregone earnings in each year. The total costs in each year are \$6,700. This stream of costs can be converted to a single figure for costs by applying the discounted present value formula. The expression for the total costs is:

$$\frac{6{,}700}{1.05} + \frac{6{,}700}{1.05^2} + \frac{6{,}700}{1.05^3} + \frac{6{,}700}{1.05^4}$$

The present discounted value of the projected benefits is calculated by the following equation. This can be evaluated pretty easily using a spreadsheet program.

$$\frac{4{,}000}{1.05^5} + \frac{4{,}000}{1.05^6} + \frac{4{,}000}{1.05^7} + \frac{4{,}000}{1.05^8} + \ldots\ldots \frac{4{,}000}{1.05^{14}}$$

One would compare the discounted present value of the costs to the discounted value of the benefits. If the former exceed the latter, do not make the investment.

See Figure 9.1 of the text for a graphical representation that corresponds to the numerical analysis shown above.

2. (LO1, 2) The basic issue in this case is that teaching a dog new tricks, which is obviously an analogy for a worker who is nearing the end of his/her working life, is a costly venture. Resources are always consumed in the process of educating and training workers. If the recipients of the human capital formation do not have a long period over which their newly obtained skills can be applied on the job, then the increase in marginal productivity that arises from the new training will not be sufficiently large to cover the initial expense incurred with the training. The costs stemming from the training cannot be covered by the benefits stemming from the training unless the amortization period is fairly long. Therefore, the statement is essentially true — it is not cost-efficient to teach old dogs new tricks.

3. (LO1, 2) In order to solve a problem like this, the first step is to list the payouts in each period for the two options with the intention of comparing them. Starting with the high school only option, in each period the payout is Y_H. For the university option, the earnings are $5,000 during the training period, but the explicit costs of the option are $5,000. This means that the net benefit, or payout, for that option in the first time period is 0. In the second time period, we are given a payout of Y_U. In order for her to attend university, the discounted present value of the net benefit of attending university (summed up over periods one and two) has to exceed the discounted present value of the net benefit of attending high school only (summed up over periods one and two). The mathematical expression is:

$$Y_H + \frac{Y_H}{1+r} < \frac{Y_U}{1+r}$$

Multiplying the first term by (1 + r)/(1 + r), we obtain a common denominator. Since this denominator must be positive, the sign does not change direction, and we obtain the following inequality:

$$Y_H\,(1+r) + Y_H < Y_U$$

which simplifies to the sought-for result. The higher the interest rate, the less likely she is to select the university option, ceteris paribus, because the higher income levels that she would earn in the future are more steeply discounted. In other words, the higher the interest rate, the more valuable the option of receiving money today relative to the option of receiving money tomorrow. Given a high interest rate, not going to university is the option that pays more money early in the time frame, and also costs less at the beginning.

4. (LO4). Note that beta is the increment in earnings that stems from obtaining more units of schooling. The true model with the subscripts is based on data observed at the level of the individual. It applies to all individuals, whether they are in Madeleine's group 1 or group 2. In theory, the average, or the expected value, of the epsilon term should be 0.

   a) This estimate is based on two points: the average earnings coupled with the average level of schooling for the higher education group, and the average earnings coupled with the average level of schooling for the lower education group. If you were to plot those two points, connect them with a line segment, and take the slope, you would obtain beta-hat. This estimate is likely to be very biased unless the average value of the epsilon term (which reflects all of the other factors that affect earnings) is equal across these two groups. That condition is unlikely to hold.

   b) The poorer families face higher interest rates. As these rates increase, the cost of obtaining a university degree increases, and thus these families are less likely to be able to send their children to university. A lot of this effect attributable to the interest rate, however, might be captured in the family income variable in her estimating equation. The implication for our purposes is that the higher the family income, the more likely the individual is to obtain a university degree, and vice versa. If the family income variable is excluded from the regression equation, an important variable determining whether or not the individual obtains a university degree is omitted from the analysis, and the influences are necessarily pushed into the epsilon terms. The returns to education, estimated as beta-hat, are likely to be overstated. In terms of the equations, epsilon-bar for the high earnings group is likely to be higher than epsilon-bar for the low earnings group, and making beta-hat exceed the true value of beta. This is because the higher earnings individuals were more likely to come from higher income families, and their higher level of income (Y-bar) is reflecting their greater educational opportunities, reflected in the epsilon term, as well as their higher level of education, reflected in the beta-hat term.

Only in the case where all individuals have a more or less equal opportunity to earn a higher degree will her approach be valid. We call this process "random assignment" of the university degree. If it does not hold, the true rate of return to education cannot be estimated accurately. This scenario is quite similar to the genetic ability issue mentioned in the textbook. Aside from genetic ability, the other factor determining earnings is family income of the individual.

5. (LO1, 2) As in problem 3, the first step is to list the payouts in each period for the three options. Yun is currently in time period zero

	time period 1	time period 2	time period 3
A) hotel	20,000	20,000	20,000
B) human resources	-5,000	50,000	50,000
C) art history	-10,000	-10,000	90,000

The next step is to calculate the discounted present value of each of these three options. In order to do so, an interest rate of 10% is assumed, but any other positive value for the interest rate could be assumed. Notice that the denominator is the same for each case. For option A, we have:

$$\frac{20{,}000}{1.1}+\frac{20{,}000}{1.1^2}+\frac{20{,}000}{1.1^3}=51{,}062$$

For option B, we have:

$$\frac{-5{,}000}{1.1}+\frac{50{,}000}{1.1^2}+\frac{50{,}000}{1.1^3}=77{,}655$$

For option C, we have:

$$\frac{-10{,}000}{1.1}+\frac{-10{,}000}{1.1^2}+\frac{90{,}000}{1.1^3}=56{,}270$$

a) In purely pecuniary terms, he should select option B.

b) The implicit consumption value that he places on that career option is its value ($56,270) relative to the value of the highest forgone alternative, which is $77,655.

c) The offspring of wealthy parents, or those with outside income, are more likely to be able to afford the high opportunity cost of option C.

d) If society deems that there is a high social value for the services of art historians, and this social value exceeds the private value, then the market is probably failing to produce a sufficient number of them, and a case might be made for subsidized tuition. Note, however, that it is costly for either the private sector or the public sector to train them due to the length of the training process, and this training process has an opportunity cost. On the other hand, if the private value coincides with the social value, we would expect those with the strongest preferences for studying art history would be willing to accept the relatively low value of that option. For option C, the low relative value is not due to really low salaries, but to the fact that the training period is relatively long. Generally, subsidized tuition would not be warranted, as the marginal cost of educating one more art historian might well exceed his/her marginal productivity as an art historian.

6. (LO3) See Figure 9.3 of the text. Recall that in this model, educational attainment has no inherent value for the workers or for the employer, as it does not affect the workers' marginal products. The productivity of the worker depends only on whether he/she is a high-ability worker (a star) or a low-ability worker (a bum).

a) In this case no workers receive the education. Since there are equal numbers of high- and low-ability workers, the average marginal revenue product is $40,000. In order to break even, firms can pay everyone a wage of $40,000. The employers will earn money by employing half of them and lose money by employing the other half. After sufficient time has passed such that the types of the workers are known, the low-ability workers might be laid off as they are not earning their marginal revenue products.

b) For low-ability workers, S = 0 and wages = $ 30,000, while for high-ability workers, S = 31,000 and wages = $50,000. The slope of the cost for the high-ability workers is 1/2, while it is unity for the low-ability workers. The low-ability workers find it optimal to receive no education, as the cost that they would have to pay ($31,000) exceeds the addition in salary that they would receive ($20,000 = $50,000 – $30,000). The high-ability workers do find it optimal to receive the education, as the minimum cost that they would have to pay, $31,000/2 = $15,500, is less than the additional income that they would receive, $20,000.

c) In this case, low-ability workers will get an education, and they will be paid $50,000, but they will only be worth $ 30,000 to their employers. The low-ability workers now find it optimal to receive education, as the cost that they would have to pay ($ 31,000/2 = $15,500) is less than the addition in salary that they would receive ($20,000 = $50,000 – $30,000). They are in the exact same situation that the high-ability workers were in before the subsidy was created. This is an unsustainable equilibrium, as employers are incurring a loss of $20,000 on every low-ability employee that they have (they are paying them the full rate), and only breaking even on every high-ability employee that they have.

7. (LO5)

a) No, this statement is fallacious, because it offers no estimate of the counterfactual earnings, i.e., what earnings would be had the women not participated in the training program. As it stands, this very simple and naïve analysis takes no account of any of the other variables that could have been acting over this period to influence the earnings of this group. The ceteris paribus condition is not satisfied. For program evaluation purposes, this is typically done by comparing the wage growth rate of the treatment group with that of a control group. The control group has to be selected such that it resembles the treatment group as closely as possible when it comes to education and training levels, innate ability, motivation and other factors that affect marginal productivity. Unfortunately, some of these traits are difficult to observe, as workers typically do not wear signs indicating that they are lazy, conscientious, intelligent, dumb, strong, weak, cooperative, abrasive, etc.

b) Yes, using this information would constitute a small step in the right direction. Over the period that the program was operating, the rate of wage growth for full-year, full-time female workers was +8.6%. While this group is not an appropriate control group (because the women in this group are not disadvantaged and are well integrated into the labour force; i.e., the training program is not designed for women with their work experiences), the approach is valid. We want to estimate what the wage growth rate would have been for the disadvantaged women had they not participated in the program. 8.6% might be interpreted as a reasonable estimate of an upper bound for that figure. The researcher obtains an increase of 50% for annual earnings for the treatment group, which far exceeds the rate of wage growth for working women on average. That probably is meaningful, although one should also take account of the number of hours that the women in the treatment group worked before and after going through the training program.

8 (LO1, 2) In order to respond completely to this question, we need to know the terms of the loans, the interest rate, and the length of the working career, but none of these pieces of information are provided. Nevertheless, in order to compare these two scenarios, as long as these parameters are the same for each of the two loan schemes, the net benefits of the two scenarios can be compared. The comparison will be restricted to the annual cost and the annual benefit, and we will abstract from the question of the time horizon. This means that we will not do any discounted present value analysis, which is somewhat immaterial to the problem anyway.

(i) If she obtains the good job from the time of graduation until retirement, the annual benefit is $10,000. If she obtains the bad job from the time of graduation, the annual benefit is $0. The expected benefit is simply the average of these two figures, since each scenario has a 0.5 probability of occurring. Her annual expected payoff would be $5,000. The cost of repaying the conventional loan depends on the interest rate and the term of the loan. We are told that she has to pay it back at a rate of $2,500 per year, but we do not know for how long. For the ICL, she will be paying an annual amount of $5,000 with a probability of 0.5 or an amount of $0 with a probability of 0.5. The expected cost is thus the average (as both outcomes are weighted equally), and it is $2,500. Regardless of which loan scheme she selects, the expected cost is the same -$2,500. Nevertheless, she would certainly prefer the outcome of obtaining the good job, as the annual net benefit in that case would be either $7,500 with the conventional loan or $5,000 with the ICL. If she were to obtain the bad job, the annual net benefit in that case would be either -$2,500 with the conventional loan or $0 with the ICL. If she selects the ICL, the annual net benefit in that case would be either $5,000 with the good job or $0 with the bad job, for an average payout of $2,500. If she selects the conventional loan, the annual net benefit in that case would be either $7,500 with the good job or -$2,500 with the bad job, for an average payout of $2,500.

(ii) If she is risk-averse, this means essentially that she values peace of mind, and is willing to sacrifice some of the average benefit in order to protect herself from the worst possible outcome. In other words, she would prefer a lottery that involves a low likelihood of getting the worst job and a lower payoff for the high wage job to the original lottery, even if the expected payoff of the original lottery is higher. She is willing to accept a lower expected payout in order to avoid the worst case scenario. Consider the unfavourable outcome of obtaining the bad job. With the ICL, the worst possible payoff is $ 0, which dominates the worst possible outcome given the conventional loan scheme, which is -$2,500. She is likely to prefer the ICL, even if it comes with a higher interest rate (which is likely), as much of the risk is shifted to the lender. If only a standard loan is available, the inherent risks of the low-wage outcome occurring and leaving her with a debt-repayment obligation, could well constitute a barrier.